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Todo sistema ortogonal es linealmente independiente

Proposición 1. *Sea H un espacio prehilbert y sea S un sistema ortogonal*

$$\implies S \text{ es linealmente independiente.}$$

***Demostración*:** Sean $\{\alpha_i\}_{i\in\mathbb{N}_n} \subset \mathbb{K}$ y $\{x_i\}_{i\in\mathbb{N}_n} \subset S$ tales que $\sum_{i=1}^{n} \alpha_i x_i = 0$. Entonces,

$$0 = \left\| \sum_{i=1}^{n} \alpha_i x_i \right\|^2 \overset{\substack{\text{Pitágoras}\\ \downarrow}}{=} \sum_{i\in\mathbb{N}_n} \|\alpha_i x_i\|^2 = \sum_{i=1}^{n} |\alpha_i|^2 \|x_i\|^2 .$$

Como $\forall i \in \mathbb{N}_n : \|x_i\|^2 > 0$, $\forall i \in \mathbb{N}_n : \alpha_i = 0$, luego S es linealmente independiente. ■

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